

Fig. 26.2. A 40-kilogram mirror being prepared for installation in LIGO. Its location fluctuates, quantum mechanically, very, very slightly: one ten-billionth the diameter of an atom.

Because objects of human size and larger have only minuscule quantum fluctuations, physicists almost always ignore those fluctuations. Discarding the fluctuations, in our mathematics, simplifies the laws of physics.

If we begin with the ordinary quantum laws that ignore gravity and then discard the fluctuations, we obtain the Newtonian laws of physics —the laws used for the past few centuries to describe planets, stars, bridges, and marbles. See Chapter 3.

If we begin with the ill-understood laws of quantum gravity and then discard the fluctuations, we must obtain Einstein's well-understood relativistic laws of physics. The fluctuations we discard are, for example, a froth of fluctuating, exquisitely tiny wormholes ("quantum foam" that pervades all of space; Figure 26.3 and Chapter 14).[46] With